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Worse than Measurement Error:

Consequences of Inappropriate Latent Variable Measurement Models

Mijke Rhemtulla

Department of Psychology, University of California, Davis

Riet van Bork and Denny Borsboom

Department of Psychological Methods, University of Amsterdam

Author Note. This research was supported in part by grants FP7-PEOPLE-2013-CIG-631145 (Mijke Rhemtulla) and Consolidator Grant, No. 647209 (Denny Borsboom) from the European Research Council. Parts of this work were presented at the 30th Annual Convention of the Association for Psychological Science. Address correspondence to Mijke Rhemtulla, Department of Psychology, University of California, Davis, One Shields Avenue, Davis, CA 95616. mrhemtulla@ucdavis.edu.

Abstract

Previous research and methodological advice has focussed on the importance of accounting for measurement error in psychological data. That perspective assumes that psychological variables conform to a common factor model, such that they consist of construct variance plus error. In this paper, we explore what happens when a set of items that are not generated from a common factor construct model are nonetheless modeled as reflecting a common factor. Through a series of hypothetical examples and an empirical re-analysis, we show that (1) common factor models tend to produce extremely biased and highly variable structural parameter estimates when the population model is not a common factor model; (2) model fit is a poor indicator of the degree of bias; and (3) composite models are sometimes more reliable than common factor models under alternative measurement structures, though they also lead to unacceptably bad solutions in some cases.

Keywords: Latent Variables; Measurement Models; Structural Equation Modeling; Measurement Error; Causal Indicators; Reflective Indicators

Over the past decades, the latent variable model has become the gold standard of measurement models in psychology (e.g., Bandalos, 2018). The model comprises a wide variety of models that range from Item Response Theory to latent class models and factor analysis (Mellenbergh, 1994). All latent variable models share the canonical assumption that observed item or test scores can be decomposed into two parts: one part that can be traced back to a psychological construct – typically represented as a latent common factor – and one part that is due to measurement error. When this assumption holds, the latent variable model is able to disentangle these two sources of variance, so that the latent variable in the model is a proxy for the psychological construct under consideration. It is widely assumed that successful application of the latent variable model supports the interpretation of test scores as measurements of the underlying construct (Borsboom, 2005, 2008; Cronbach & Meehl, 1955; Markus & Borsboom, 2013; Maul, 2017). In addition, when embedded in a larger structural equation model, latent variables allow the modeller to estimate unbiased relations among psychological constructs and model causal processes involving these constructs (Bollen & Pearl, 2013).

The most popular alternative to latent variable modeling is to treat observed composites as proxies for psychological constructs; this strategy includes the use of summed or averaged item scores obtained in the absence of a measurement model, but also the use of weighted composites like principal component scores and related methods like partial least squares regression (Hair, Ringle & Sarstedt, 2011; Rigdon, 2012). As methodologists are keen to point out, however, such composite scores conflate true construct variance and measurement error, resulting in attenuated estimates of construct relations and a host of concomitant problems (Bollen & Lennox, 1991). In the past few years, research has documented the effects of neglecting measurement error in detail; for instance, Cole & Preacher (2014) argued that the widespread neglect of measurement error in psychological science likely compromises the quality of scientific research; such concerns have led some scholars to advocate the use of latent variable models wherever possible (Borsboom, 2008).

The existing methodological literature almost uniformly assumes that some version of the latent variable model is true, and then analyses what goes wrong when the presence of measurement error is ignored (e.g., Cole & Preacher, 2014; Jaccard & Wan, 1995; Ledgerwood & Shrout, 2011; Rhemtulla, 2016). But such analyses tend to ignore the other side of the

psychometric coin. That is, very little research so far has evaluated what happens when the underlying measurement assumptions are in fact *false*; that is, when observed scores are *not* composed of underlying construct variance plus random measurement error, but are nevertheless analyzed *as if they were*. A notable exception comes from the field of marketing, where causal-indicator models are relatively more common than in psychology, and at least one simulation study has studied this question empirically (Jarvis, Mackenzie, & Podsakoff, 2003). This question is acute in light of the fact that many recent papers in psychiatry and clinical psychology have in fact argued that latent variable models have been over-applied in these fields, by applying a latent variable model to account for the correlation between symptoms that are more likely to feature direct causal relations (e.g. insomnia $\rightarrow$ fatigue in depression; panic attacks $\rightarrow$ worry about panic attacks in panic disorder; delusions $\rightarrow$ paranoia in psychosis; Borsboom & Cramer, 2013). Similar criticisms have been voiced in the context of research into attitudes (Dalege, Borsboom, van Harreveld, van den Berg, Conner, & van der Maas, 2016), quality of life (Kossakowski, Epskamp, Kieffer, van Borkulo, Rhemtulla & Borsboom, 2016), personality (Cramer et al., 2012), and intelligence (van der Maas, Dolan, Grasman, Wicherts, Huizenga, & Raijmakers, 2006; van der Maas, Kan, & Borsboom, 2014).

Terminology

Throughout this paper, we use the term “latent variable model” and “common factor model” interchangeably to refer to a unidimensional model that represents a construct as a latent variable with reflective indicators only. We use “construct” to refer to the conceptual variable that the researcher intends to represent using a statistical proxy, i.e. the attribute under investigation (Markus & Borsboom, 2013). Following Rigdon (2012, 2013), we refer to the statistical variable (whether it is a latent variable or a composite score) as a “proxy” that stands in for the construct under investigation.

Overview

The goal of the present paper is to elucidate the consequences of using common factor models in circumstances when the true model that generated a set of observed variables is something different. We begin with a theoretical overview of latent variable models that describes the problem of measurement error and how latent variable models solve this problem.

We then argue that applications of latent variable models in psychology are frequently unjustified, and we propose that this practice may lead to three kinds of consequences: construct invalidity (i.e., modeling the wrong construct), model misfit, and biased structural parameter estimates. We show analytically that when the true model is a composite—that is, the construct is an inductive summary (Cronbach & Meehl, 1955) in the sense that it is truly equivalent to a sum of a set of observed variables—rather than an underlying common factor, the bias that arises is similar in strength and opposite in direction to the bias that arises when a construct is truly equivalent to an underlying common factor that is erroneously modeled as a composite. We then present three more nuanced examples. In each example, a population model is defined in which the relations among a set of variables that represent a construct do not conform to a common factor model (Example 1 is a set of causal indicators, Example 2 is a set of MIMIC indicators, and Example 3 is a directed graph). For each example, we fit composite and reflective latent variable models to the population data and interpret the resulting discrepancies. For the sake of generalization, each example is accompanied by a “parameter sweep” in which we extend the findings from a single population to a series of similar populations with a range of parameter values. Finally, we re-analyse a published data set using three different measurement models and compare the results to demonstrate the interpretational differences that can arise due to one’s choice of measurement models.

Theoretical Background

The common factor model is based on the central equation of classical test theory that scores on an observed variable are composed of true score plus random error, $X = T + E$. In classical test theory, true score is simply defined as an individual’s average score over repeated sampling. The logic of factor analysis relies on the assumption that a set of observed variables all have the same common score (or a rescaled version of it) to isolate that shared variance and separate it from error. That is, what was true score in classical test theory becomes the common factor (Jöreskog, 1971). Each person’s score on every measured variable in the set contains their common factor score, η , weighted by a factor loading for that item, λ , plus random error: $y = \lambda\eta + \varepsilon$. In certain contexts, the notion that all test items measure exactly the same thing (e.g., a set of simple addition items are interchangeable measures of addition skill) is plausible. When measuring broader psychological constructs, however, such as intelligence, depression,

extraversion, school readiness, or aggression, the set of observed variables are often not seen as interchangeable manifestations of the construct. Thus, in the context of factor analysis, “true score” and “error” are usually recast as “common” and “unique” variance components. This formulation makes explicit that the latent factor is conceptualized as whatever a set of reflective indicators have in common, and item residuals are conceptualized as whatever is left over, or unique to each item.

When the common factor model is appropriate, it confers strong benefits to researchers who use it. By separating variance related to the construct from error variance, the model allows unbiased estimates of relations among constructs, and thus it produces accurate tests of theories about these relations. Cole and Preacher (2014) demonstrated the pernicious consequences that can arise from modeling a construct as a composite (i.e., a sum of observed variables) instead of a latent variable when the latent variable model is appropriate: Measurement error in reflective indicators is retained in the composite score, inflating its variance and thereby deflating its direct relations with other variables. These deflated bivariate paths, in turn, are too low to fully account for covariances between pairs of variables that are only indirectly connected via these deflated paths. This under-estimation puts pressure on other parts of the model to compensate by over-estimating other paths, and can cause spurious model misfit by putting pressure on truly-zero paths to be estimated as non-zero. These problems are exacerbated in complex models, where bias in one part of the model propagates to other parts, producing substantially over- and under-estimated parameter estimates and the appearance of either better or worse fit than would occur if the construct measures were free of error. For these reasons, increasing numbers of psychological researchers are choosing to use latent variable SEM to model their constructs of interest.

But the common factor model carries two strong implications for the relations among the set of observed indicators, and for the relations between those indicators and the construct they measure. First, the factor model implies that the *entirety* of the true factor score is present in every observed variable, which means that any variance not shared among *all* indicators in the set will not be part of the modeled latent variable. This implication leads to problems whenever items are selected to capture disparate *facets* of a construct’s variance, because the common factor will not capture these unique facets (Bagozzi, 2007). Second, the factor model implies that

all pairs of observed variables are uncorrelated after conditioning on the latent variable, an assumption known as *local independence*. If a subset of variables share unique variance, due to either a direct causal relation between observed indicators or due to an unspecified common cause, in many cases local independence will be violated, rendering the factor model misspecified¹ (Epskamp, Rhemtulla, & Borsboom, 2017).

Cole & Preacher's (2014) argument presupposes that most measures of psychological constructs *ought to be modeled* as reflective indicators of a latent variable. This supposition appears to be quite common. It is typical for researchers to use latent variable SEM without justifying the appropriateness of the common factor model for the constructs under investigation. But while the common factor model may be the gold standard in terms of psychometric measurement models, it does not follow that any set of observed variables related to a construct should conform to it. A construct may instead be composed of or caused by a set of variables that are measured by items (as represented by composite, causal indicator, and MIMIC models), or it may emerge from a network of directly interrelated items (as represented by a network model). We detail these alternatives in the next section.

Common Factor Models are Used by Default in Psychology

Anecdotally, we have noticed a strong tendency in Psychology for latent variable models to be employed without an attempt to justify the choice of measurement model. A perusal of the most recent issue of several major APA journals supports this contention: In the following paragraphs we describe six examples taken from the most recent issue of each of six APA journals at the time of writing this (August 2016):

In *Developmental Psychology*, Yan and Dix (2016) modeled children's Mastery Motivation as a latent variable indicated by observer ratings of children's enthusiasm and persistence when interacting with a toy. To justify the model, the authors stated that these two indicators were correlated and that they were "hypothesized to form the latent construct". In

¹ While it is usually possible to model some residual covariance to account for these extra-factor relations, doing so assumes that the correlated residual variance is due to something external to the construct. When the shared variance is potentially relevant to the construct (e.g., it arises because two items measure the same facet of a construct), modeling a residual covariance removes that variance from the latent variable.

Health Psychology, Hoerger, Scherer, and Fagerlin (2016) modeled the construct Behavioral Preferences and Intentions as a latent variable indicated by women's self-rated preference for taking chemoprevention, as well as their intentions of talking to their doctor, looking up more information, and taking a medication. This choice of model was explicitly atheoretical, as the authors explained that the indicators were combined to simplify interpretation. In the *Journal of Educational Psychology*, Lee and Bull (2016) modeled Working Memory and Updating as a latent variable indicated by two recall task scores and an updating task score. Despite these tasks representing two distinct constructs (i.e., the authors claim to be interested in assessing the relation between working memory and updating), they were modeled as indicators of a single latent variable "to obtain separate estimates of true scores and measurement error" (p. 873). In the *Journal of Abnormal Psychology*, Davies, Hentges, Coe, Martin, Sturge-Apple, and Cummings (2016) modeled Emotional Insecurity as a latent variable indicated by children's displays of distress, vigilance, avoidance, disengagement, and secure behaviors in the face of parental conflict. In the *Journal of Family Psychology*, Vieira, Matias, Ferreira, Lopez, and Matos (2016) modeled Work-Family Dynamics as a latent variable indicated by subscale scores on work-family conflicts of time, emotional strain, and behavior. Finally, in the *Journal of Personality and Social Psychology*, Kandler, Riemann, Angleitner, Borkenau, Spinath, and Penke (2016) investigated the structure of Creativity by comparing unidimensional, bifactor, and higher-order common factor models indicated by self-reported creativity, peer-reported creativity, direct observation of creative behavior, and ratings of originality and elaboration based on a line-drawing completion test.

Crucially, in none of these examples is it necessarily obvious that a common factor measurement model is the most appropriate model for the construct. In the first three studies cited above, the modeled common factors are explicitly intended to combine distinct constructs: Children's enthusiasm and persistence are understood as distinct, non-interchangeable variables that together "form the construct", in the authors' words, rather than the other way around; Behavioral Preferences and Intentions is explicitly a combination of two discrete constructs rather than a pair of indicators of an underlying construct; and Working Memory and Updating are also explicitly discussed as two distinct but related constructs. The next three examples are less explicit, but in each study a case could be made for a non-common-factor construct model: Children's insecure behaviors may behave in a compensatory way, such that certain behaviors

(e.g., vigilance) make others less likely (e.g., avoidance), given a fixed level of insecurity; work-induced strains on time, emotional energy, and behavior could be argued to be more likely to contribute to Work Family Dynamics than the other way around; and Creativity may plausibly be described as a dynamic system in which self-perceptions of creativity, peer-perceptions of creativity, and creative behaviors influence each other, with the construct emerging from this network.

While these examples are just that, and thus they do not provide a complete or necessarily representative review of the literature, the fact that each one of these articles chosen from the single most recent issue of six major APA journal contains an example of a common factor model that is not well justified supports our observation that researchers and reviewers use and accept common factor models as a working assumption, without detailed consideration of alternatives (see Burke Jarvis, MacKenzie, & Podsakoff, 2003 for a more thorough review in the field of Marketing that leads to the same conclusion). In light of this state of affairs, it is crucial that we understand the consequences of such practices.

Three Possible Consequences of an Incorrect Measurement Model

In a structural equation model, the structural model aims to describe relations among latent variables, while measurement models relate these latent variables to the data. When the measurement model is misspecified, it can lead to three types of problems. First, the latent variable will not represent the construct the researcher intends it to represent, posing a threat to construct validity. Consequently, estimated relations among latent variables (i.e., the structural model) can be poor estimates of the relations among the constructs that a researcher intends to represent. Third, these problems at the measurement and structural levels can lead to poor fit which may or may not reflect problems with the theory that the broader model is meant to test.

To see what can go wrong, suppose a researcher has measured income, education, and occupational prestige as indicators of the target construct *socioeconomic status* (SES). SES is frequently used as an example of a variable that is better represented as a composite or effect of its components (e.g., income, education, occupational prestige) than as an underlying common cause (Bollen & Lennox, 1991; Hauser & Goldberger, 1971). Modeling these indicators as reflective indicators of a common factor captures only the shared variance among them. That is,

a person with a high score on the latent variable is expected to have high income, high education, and high occupational prestige. A person with high education but low income and low occupational prestige (e.g., a graduate student) would have her aberrant education status attributed to the residual variance of the education indicator; that is, her high education would be interpreted as unrelated to SES. Similarly, an unemployed wealthy heir with high income, no education, and no occupational prestige would have his wealth attributed to factors unrelated to SES. In other words, to the extent that income, education, and occupational prestige diverge from each other, these differences are modeled as residual variance, orthogonal to the construct. This shared-variance version of SES is not a good proxy for the target SES construct, which is more likely to include unique contributions from education, income, and occupation. Thus, applying a latent variable model inappropriately is a threat to construct validity (Burt, 1976).

When an inappropriate measurement model is embedded into a larger structural model, it can lead to model misfit and can change the meaning of the latent variable by way of shifting the factor loadings. The common factor model assumes that all relations between indicators of a latent variable and other variables in the model are mediated by the latent variable. For the model to fit well, these relations must thus be reproducible as a function of the estimated structural model coefficients and the factor loadings. Suppose that a researcher wants to know the extent to which maternal IQ predicts children's SES (see Figure 1). The common factor model requires that the effect of maternal IQ on children's income, education, and occupational prestige is fully mediated by the latent SES variable. For the model to fit, these effects must be proportional to the factor loadings of these indicators on the latent SES variable (i.e., it must be a stronger predictor of those indicators with higher factor loadings). To the extent that this proportionality constraint is not met, the model will be unable to account for the association between maternal IQ and the indicators of SES, leading to statistical misfit. Even more problematically, the estimation algorithm will seek to minimize this misfit by selecting measurement model parameters (factor loadings) that maximally account for the covariances between maternal IQ and the indicators of SES, potentially dramatically re-defining the latent variable in the process.

Finally, whether or not the proportionality constraints are met, relations between the misspecified construct proxy and other variables in the model will not be accurate estimates of relations with the target construct. Continuing the example from the previous paragraphs, now

suppose that the proportionality constraint is met, and that the statistical model fits well. Because maternal IQ affects all the indicators, it contributes to the shared variance among the indicators (i.e., all indicators share some variance due to a shared influence of maternal IQ). The structural path coefficient from maternal IQ to the latent variable represents the proportion of the latent variable's variance that is shared with maternal IQ. Because variance that is unique to each indicator is not considered to be part of the latent variable, it is not involved in the estimation equations for this proportion. If the target construct includes not only shared aspects but also unique aspects of the indicators, then the estimated structural regression coefficient will overestimate the proportion of SES that is accounted for by maternal IQ. To take an extreme example, if the three indicators of SES share very little variance, while almost all that shared variance is due to maternal IQ, the model will produce low factor loadings (because a relatively minor proportion of the indicators' variance is shared) and a very high structural path coefficient (because most of that shared variance will be attributed as due to maternal IQ). This result would be interpreted as a strong predictive effect of maternal IQ on SES. Figure 1 depicts these three consequences of inappropriate latent variable modeling. In the next section, we derive analytically the degree of overestimation that arises in the specific case of truly composite indicators that are modeled as reflective.

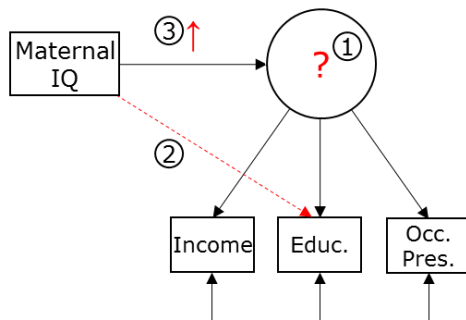


Figure 1. Three Consequences of an Inappropriate Common Factor Model. (1) The latent variable reflects the shared variance among a set of indicators, which is not the same thing as the construct of interest. (2) If the effects of the predictor on the indicators are not proportional to the factor loadings, the model will not fit (direct paths to indicators may be necessary to achieve model fit). (3) Structural path coefficients estimate the association between other variables in the model and the shared variance among indicators, rather than the association between other variables in the model and the target construct variance.

Bivariate Structural Correlations are Overestimated when the True Model is Composite

In the previous section we described a situation in which structural parameters will be over-estimated when the true model includes composite indicators that are erroneously modeled as reflective. This is just one possible way in which a measurement model might be misspecified, and it is likely too simplistic to be true for many, if any, psychological constructs. Nonetheless, it is instructive to see how and why this positive bias emerges. We first show this in the simplest case, when the reflective latent variable model, though it is not the true model, nevertheless perfectly reproduces the population covariance matrix. Let M be a unit-weighted composite of p components, $m_1 \dots m_p$, such that $M = \sum_{i=1}^p m_i$. Let Y be a variable with unit variance and

$\text{cov}(m_i, Y) = b$ for all m_i . For the sake of simplicity, assume all m_i have unit variance and $\text{cov}(m_i, m_j) = c$ for all $m_i \neq m_j$. The correlation between M and Y is thus:

$$\begin{aligned}
 \text{cor}(M, Y) &= \frac{\text{cov}(M, Y)}{\sqrt{\text{var}(M) \text{var}(Y)}} \\
 &= \frac{\text{cov}(\sum m_i, Y)}{\sqrt{\text{var}(\sum m_i)(1)}} \\
 &= \frac{p \text{cov}(m_i, Y)}{\sqrt{p + p(p-1)c}} \\
 &= \frac{pb}{\sqrt{p + p(p-1)c}}
 \end{aligned} \tag{1}$$

When a reflective indicator model is imposed on the set of components m_i such that $m_i = \lambda_i M^* + \varepsilon_i$, this model will perfectly reproduce the covariance matrix Σ_m when $\lambda_i = 1$, $\text{var}(M^*) = c$, and $\text{var}(\varepsilon_i) = 1 - c$. The value of $\text{cov}(M^*, Y)$ that reproduces the covariances between each component m_i and Y is b . The correlation between M^* and Y then becomes:

$$\text{cor}(M^*, Y) = \frac{\text{cov}(M^*, Y)}{\sqrt{\text{var}(M^*) \text{var}(Y)}} = \frac{b}{\sqrt{c}} \tag{2}$$

By dividing Equation (1) by Equation (2), we get an inflation factor that shows how much bigger the latent variable model correlation is compared to the composite variable correlation:

$$\frac{\text{cor}(M^*, Y)}{\text{cor}(M, Y)} = \frac{b}{\sqrt{c}} \cdot \frac{\sqrt{p + p(p-1)c}}{pb} = \sqrt{\frac{1 + (p-1)c}{pc}} \quad (3)$$

This inflation factor is a function of the number of components, p , and the covariance among the components, c . Figure 2 (top left) displays the degree of over-estimation as a function of c (along the x-axis) and p (separate lines).

This result is precisely the opposite of the classic attenuation due to measurement error effect (Spearman, 1904). Whereas the attenuation effect is that using error-laden variables (e.g., a sum score based on a set of congeneric items) rather than reliable variables (e.g., a latent variable indicated by a set of congeneric items) results in under-estimation of correlations, we have just shown that using a latent variable measurement model when a sum score is true results in *misdisattenuation*: Correlations are disattenuated of measurement error that does not exist, producing over-estimation bias.

We can extend this finding to the case in which the components are measured with error. Let $x_1 \dots x_p$ be unreliable indicators of $m_1 \dots m_p$, such that $x_i = \sqrt{\rho}m_i + e_i$ and $\text{var}(e_i) = 1 - \rho$. Thus, ρ is the reliability of each unit-variance indicator. Let $X = \sum_{i=1}^p x_i$ be the composite of these unreliable observed variables. The correlation between X and Y is thus:

$$\begin{aligned} \text{cor}(X, Y) &= \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}} \\ &= \frac{\text{cov}(\sum x_i, Y)}{\sqrt{\text{var}(\sum x_i)(1)}} \\ &= \frac{p \text{cov}(x_i, Y)}{\sqrt{p + p(p-1)c\rho}} \\ &= \frac{pb\sqrt{\rho}}{\sqrt{p + p(p-1)c\rho}} \end{aligned} \quad (4)$$

When a reflective indicator model is imposed on the set of observed variables x_i such that $x_i = \lambda_i M^{**} + \varepsilon_i$, this model will perfectly reproduce the covariance matrix Σ_m when $\lambda_i = 1$, $\text{var}(M^*) = c\rho$, and $\text{var}(\varepsilon_i) = 1 - c\rho$. The value of $\text{cov}(M^{**}, Y)$ that reproduces the covariances between each component x_i and Y is $b\sqrt{\rho}$. The correlation between M^{**} and Y then becomes:

$$\text{cor}(M^{**}, Y) = \frac{\text{cov}(M^{**}, Y)}{\sqrt{\text{var}(M^{**}) \text{var}(Y)}} = \frac{b\sqrt{\rho}}{\sqrt{\rho c}} = \frac{b}{\sqrt{c}} \quad (5)$$

Note that item reliability drops out of this equation, making it identical to Equation (2). That is, $\text{cor}(M^{**}, Y) = \text{cor}(M^*, Y)$. In contrast to its effect on the observed composite score, measurement error does not affect structural correlations in the latent variable model.

By dividing Equation (1) by Equation (4), we get a multiplier that shows how much smaller the unreliable composite correlation is compared to the perfectly reliable composite variable correlation:

$$\begin{aligned} \frac{\text{cor}(X, Y)}{\text{cor}(M, Y)} &= \frac{bp\sqrt{\rho}}{\sqrt{p + p(p-1)c\rho}} \cdot \frac{\sqrt{p + p(p-1)c}}{pb} \\ &= \sqrt{\frac{\rho[p + p(p-1)c]}{p + p(p-1)c\rho}} \\ &= \sqrt{\frac{\rho + (p-1)c\rho}{1 + (p-1)c\rho}} \end{aligned} \quad (6)$$

This inflation factor is a function of the number of components, p , the covariance among the (perfectly reliable) components, c , and the reliability, ρ . Figure 2 (right column) displays the degree of under-estimation as a function of c and p (top-right, holding ρ constant at .8) and ρ and p (bottom-right, holding c constant at .5). The bottom-left figure displays the degree of over-estimation that occurs when the latent variable model is fit, as a function of ρ (x-axis; as shown in Equation (5) this over-estimation is not affected by ρ) and p (separate lines).

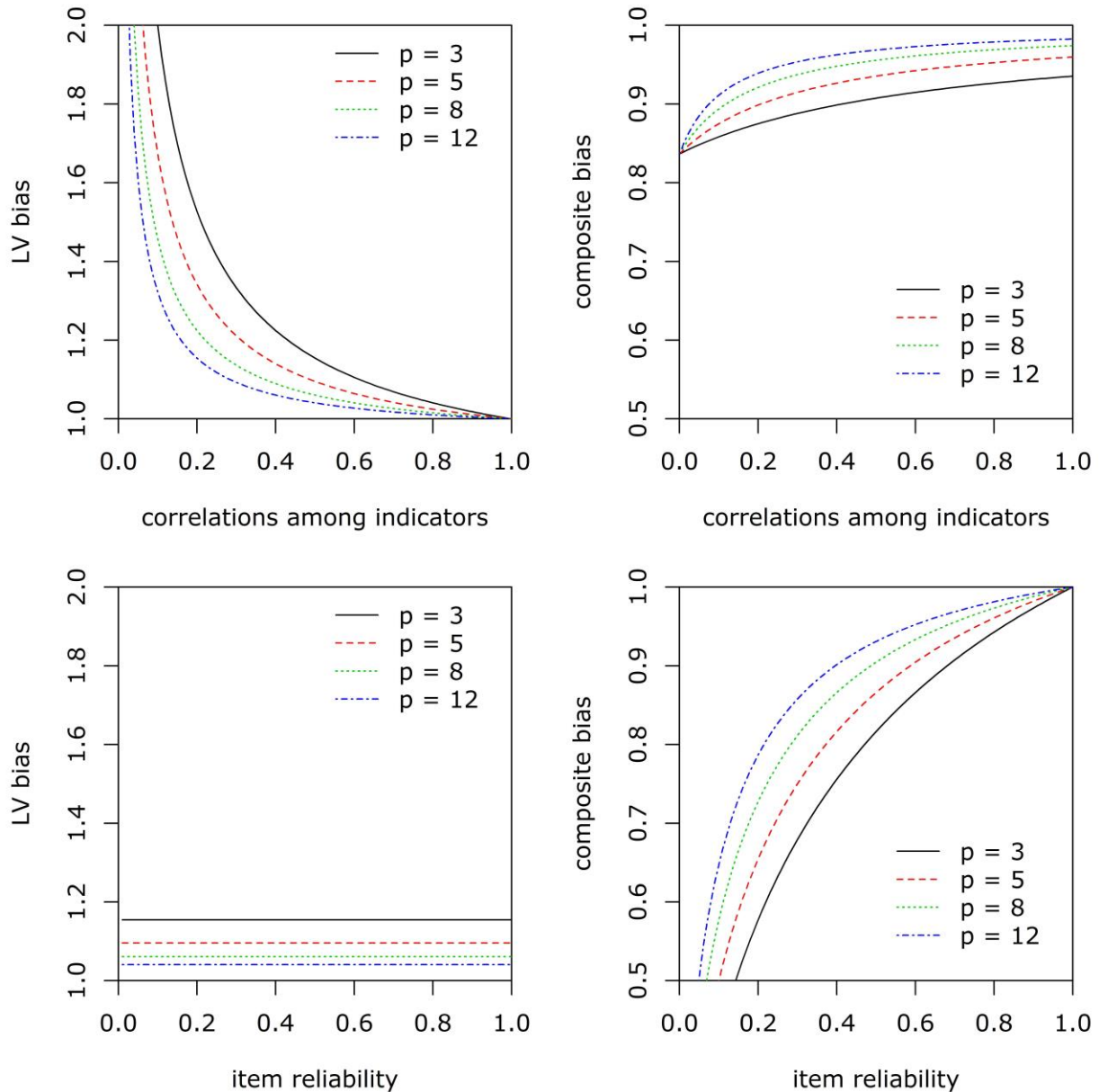
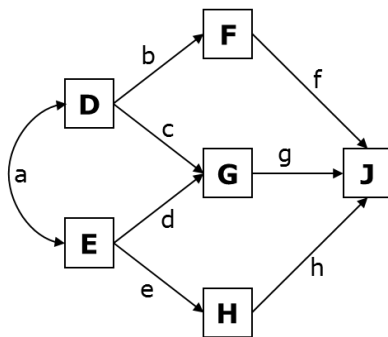


Figure 2. The factor by which a correlation coefficient is over- or under-estimated as a consequence of the model applied when the true construct is defined as a composite of variables that are perfectly measured. Left column: the inflation factor describing the over-estimation of the latent variable correlation as a function of the correlation between indicators, c , the number of indicators, p , and the reliability of the observed measures of each component, ρ . Right column: the multiplier describing the under-estimation of the observed composite correlation as a function of c , p , and ρ . Top row: ρ is held constant at .8. Bottom row: c is held constant at .5.

Cole and Preacher (2014) showed that the attenuating effects of measurement error in a single variable can “put pressure” on the model that propagates through both estimated and constrained paths to produce a combination of under- and over-estimation in a complex path model. The effect that we have described above behaves in precisely the same way, producing the opposite bias to what that paper showed. For example, we replicated Model 3c from Cole & Preacher (2014; see Figure 3), which those authors used to show how measurement error in any one variable in a 6-variable path model produced upward or downward bias in the 8 modeled paths. To replicate their finding, we replaced each variable in turn with a version of the variable that had reliability of .75, which would result from summing 3 scale items that each had .5 reliability (and were thus inter-correlated at .5). We then replaced each variable in turn with an incorrect reflective measurement model fit to 3 indicators that were inter-correlated at .5, with the important difference that the true variable was now assumed to be a sum of the three items, rather than a common cause.



path	true value	Variable with unmodeled measurement error						Variable incorrectly modeled as latent					
		D	E	F	G	H	J	D	E	F	G	H	J
a	.3	.260	.260	.300	.300	.300	.300	.343	.343	.300	.300	.300	.300
b	.3	.260	.300	.260	.300	.300	.300	.338	.300	.342	.300	.300	.300
c	.3	.254	.324	.300	.260	.300	.300	.353	.269	.300	.342	.300	.300
d	.3	.324	.254	.300	.260	.300	.300	.269	.353	.300	.342	.300	.300
e	.3	.300	.260	.300	.300	.260	.300	.300	.338	.300	.300	.342	.300
f	.3	.301	.301	.259	.309	.301	.260	.299	.299	.345	.290	.299	.346
g	.3	.301	.301	.309	.258	.309	.260	.299	.299	.289	.339	.289	.346
h	.3	.301	.301	.301	.309	.259	.260	.299	.299	.299	.290	.345	.346
pop. RMSEA		.013	.013	.010	.017	.010	.000	.012	.012	.008	.015	.008	.000

Figure 3. Replication of Model 3c from Cole & Preacher (2014). Each table column represents results from a single model in which one variable either had added unmodeled measurement error (resulting in reliability of .75 for that variable), or was a composite of 3 correlated items that was incorrectly modeled as a common factor (resulting in estimated factor loadings of $\sqrt{.5}$). Highlighted cells are underestimates; bold-faced numbers are overestimates. Pop. RMSEA is the population value of the root mean-squared error of approximation, which is a function of the minimum of the fit function and the model degrees of freedom: $popRMSEA = \sqrt{f / df}$; values closer to 0 reflect better fit.

The resulting path estimates for each model are presented in Figure 3. We found that for every path that Cole & Preacher (2014) reported to be under-estimated, the incorrect reflective

model resulted in a similar degree of over-estimation, and vice versa. We find this example particularly striking because there would be no way to distinguish the models statistically: in both cases, the 3 items measuring a construct were described by inter-item correlations of .5, resulting in a factor model with perfect local fit (i.e., perfect fit before embedding it in the larger model). The degree of model misspecification that results from both types of misspecification is very similar. In the case that those 3 items correlate at .5 because they share a common cause but contain measurement invariance, the correct course of action is to use a latent variable model. But in the case where the unique variance of each item is integral to the construct, applying a common factor model induces precisely the opposite bias that measurement error creates.

Alternative Construct Models

In the previous section, we showed that it is possible to induce roughly equal and opposite bias to the effects of measurement error by using common factor models to represent constructs that are actually perfectly represented by composite. Admittedly, though, the composite model is unlikely to ever perfectly describe psychological variables, because it requires that the components of a construct are measured without error, and that *all* components of the construct are measured. In this section, we describe three possible alternative models that relate manifest variables to each other and to the constructs they are intended to measure: causal indicator, multiple-indicator multiple-cause (MIMIC), and network models.

Causal Indicator Model. In a composite model, the statistical proxy variable of the construct is entirely determined by the observed indicators: Once the scores on the indicators are known, the value of the composite score is known as well (van der Maas, Kan, & Borsboom, 2014). A causal indicator model is similar to a composite model in that the proxy is an outcome of the set of indicators; this model differs only in that the proxy is not assumed to be fully determined by the set of observed variables, so it is latent (Bollen & Bauldry, 2011; Bollen & Lennox, 1991). Bollen & Bauldry (2011) also draw a conceptual distinction between causal and composite indicators, where causal indicators are required to display conceptual unity but composite indicators are not. We make no such distinction here, but simply use the term composite indicators when the proxy has no residual variance, and causal indicators when it does.

The latent variable, η , is defined:

$$\eta_i = \gamma' \mathbf{x}_i + \zeta_i \quad (7)$$

where γ is a $Q \times 1$ vector of indicator weights, $\mathbf{x}_i$ is person i 's vector of scores on the Q causal indicators, and ζ_i is person i 's residual score. For this and all other measurement models in this paper, we have omitted variable means and intercepts for the sake of simplicity. One such construct is SES, which is influenced by income, education, and occupational prestige.

In contrast to a composite model, the latent variable in a causal indicator model is not entirely determined by the weighted sum of the indicators. For example, it would be possible for someone with low values on all indicators to nonetheless have high SES (e.g., an Olympic swimmer). As in any regression model, the predictors (i.e, the composite indicators) are assumed to be measured without error. This assumption is likely to be violated (e.g., it is probably impossible to perfectly measure occupational prestige), which will lead to a biased representation of the relations between the construct and its predictors (Westfall & Yarkoni, 2016). In practice, that problem may be addressed by treating each item as a single indicator of a latent variable with fixed reliability (Savalei, in prep; Oberski & Satorra, 2013). The mere existence of error in the indicators does not justify using a common factor model.

A causal indicator measurement model cannot be estimated on its own because the proxy cannot be uniquely identified by a set of causes. The unknown parameters of the model include the covariances among the set of indicators, the causal indicator weights, and the residual variance of the latent variable. Of these, there is enough information in the set of measured variables to estimate the covariances among indicators, but the weights and residual variance are underdetermined (Howell, Breivik, & Wilcox, 2007). In practice, researchers typically look for two or more reflective indicators of a construct so that the model can be identified (see MIMIC models, next).

Multiple-Indicator Multiple-Cause (MIMIC) Model. A MIMIC model combines the causal indicator and common factor models (Jöreskog & Goldberger, 1975; Zellner, 1970). The latent variable is identified by a set of reflective indicators and it is additionally influenced by a set of causal indicators. The latent variable is defined as a function of Q causal indicators and a

residual, as in Equation (7), and the P reflective indicators are defined by a set of P equations of the form:

$$y_{pi} = \lambda_p \eta_i + \varepsilon_{pi}, \quad (8)$$

where y_{pi} is person i 's score on the p^{th} reflective indicator of the latent variable, λ_p is that indicator's factor loading, and ε_{pi} is its residual. An example of a when this model is appropriate is if the construct Healthy Lifestyle were indicated by the causal indicators weekly exercise hours, vegetable intake, and hours of sleep, and by reflective indicators that ask participants and their peers to rate the perceived health of the participants' lifestyle. In this model, the shared variance among the reflective indicators (e.g., self- and peer-ratings of healthy lifestyle) is identified with the proxy, such that Healthy Lifestyle is modeled as whatever those self- and peer-ratings have in common. The estimated causal indicator weights reflect the extent to which the causal indicators predict this shared variance. The factor residual, ζ , comprises the shared variance among the reflective indicators that is not predicted by the causal indicators.

Network Model. Network models are a relatively recent addition to the class of construct models in psychology. In these models, there is no proxy variable that represents the construct; instead, indicators are allowed to relate to each other directly, forming a dynamic system (Borsboom & Cramer, 2013; Cramer & Borsboom, 2015; Cramer, Waldorp, van der Maas, & Borsboom, 2010; Cramer et al., 2012; Schmittmann, Cramer, Waldorp, Epskamp, Kievit, & Borsboom, 2011). The motivating idea behind network models is that many psychological constructs are better understood as systems of interacting components, rather than as unidimensional unobservable variables. For simplicity, in the present paper we consider only directed acyclic networks, in which observed variables have direct effects on each other, but there are no feedback loops among them. In terms more familiar to structural equation modeling, these are simply recursive path models. The model can be described by the equation

$$\mathbf{x}_i = \mathbf{B}\mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad (9)$$

where $\mathbf{x}_i$ is person i 's vector of scores on the observed variables, $\mathbf{B}$ is a $Q \times Q$ lower-triangular matrix of regression weights that relate the observed variables to each other, and $\boldsymbol{\varepsilon}_i$ is

person i 's vector of residual scores. Although it may seem strange to view a recursive model as a measurement model, Equation (9) in fact has a long history in psychometrics under the header of image factor analysis (see Guttman, 1953), and recent research has suggested that many psychological disorders may be best characterized by this type of a model, in which symptoms influence each other directly (e.g., lack of sleep leads to fatigue) and the construct simply is the network of mutually reinforcing symptoms, rather than an underlying cause of the symptoms (Borsboom & Cramer, 2013; Cramer & Borsboom, 2015; van de Leemput et al., 2014).

In the next 3 sections, we show by example what can happen when each of these types of true models is fit using composite and reflective models. We use parameter sweeps to generalize our results to a fuller set of parameter values within a single model, to give a sense of the generalizability of the results.

Example 1: Causal Indicators

Example 1 is a mediation model involving three variables: an exogenous predictor, X , a mediator M , and an outcome variable, Y . The population model is depicted in Figure 4. The construct represented by M is caused by 4 imperfectly-measured causal indicators, $m_1 - m_4$, which fully mediate the effect of X on M . That is, X independently effects each of the causal indicators of M , which in turn cause variation in M . For example, suppose the effect of parental permissiveness on teen sexual activity is mediated by exposure to sexual media content. Such an effect would imply that permissiveness increases teens' exposure to one or more types of sexual media, the composition of which leads to higher levels of sexual activity.

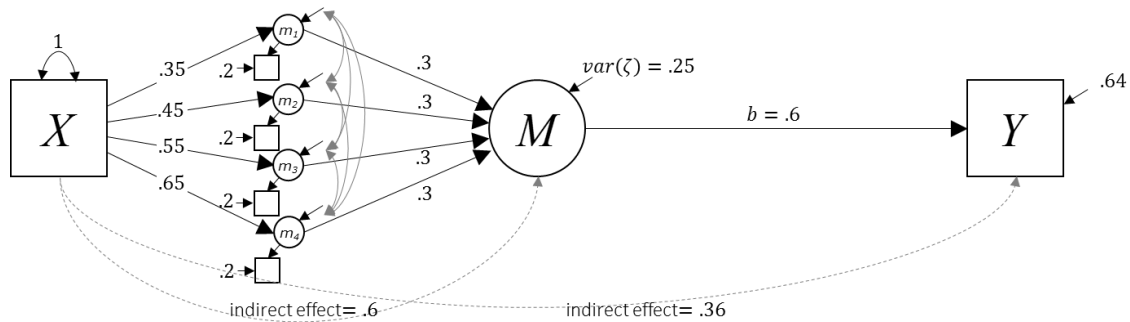


Figure 4. Example 1 true model. The residual correlation between each pair of latent causal indicators ($m_1 - m_4$) is fixed such that the total correlation between each pair of latent causal indicators is equal (total covariance = .36). The total variance of each

latent and observed variable in the model is 1, with the exception of residuals. The total indirect effect of X on M via $m_1 - m_4$, (depicted by the dotted line) is .6, and the indirect effect of X on Y is .36.

To simplify the model somewhat, we have set all causal indicator paths to be equivalent, as well as setting the covariance between each pair of indicator variables to be equivalent. Each causal indicator is measured with a single variable with reliability of .8. This measure differs from the simple composite model that we considered earlier because we have introduced residual variance in M (i.e., it is no longer a simple composite), in addition to error variance in each causal indicator. The true total effect of X on M is equal to .6, and the effect of M on Y is also .6. It is important to note that the true model cannot be estimated from population data, as it is not identified for two reasons: 1) the residual variance of each measured causal indicator could not be estimated, and 2) the residual variance of the mediator M could not be identified given the information in the observed data. However, it is possible to fit a composite model by summing up the 4 observed indicators of M , and a latent variable model using those 4 indicators of M as reflective indicators.

Figure 5 depicts the results of six models that were fit to the population covariance matrix implied by the model in Figure 4.

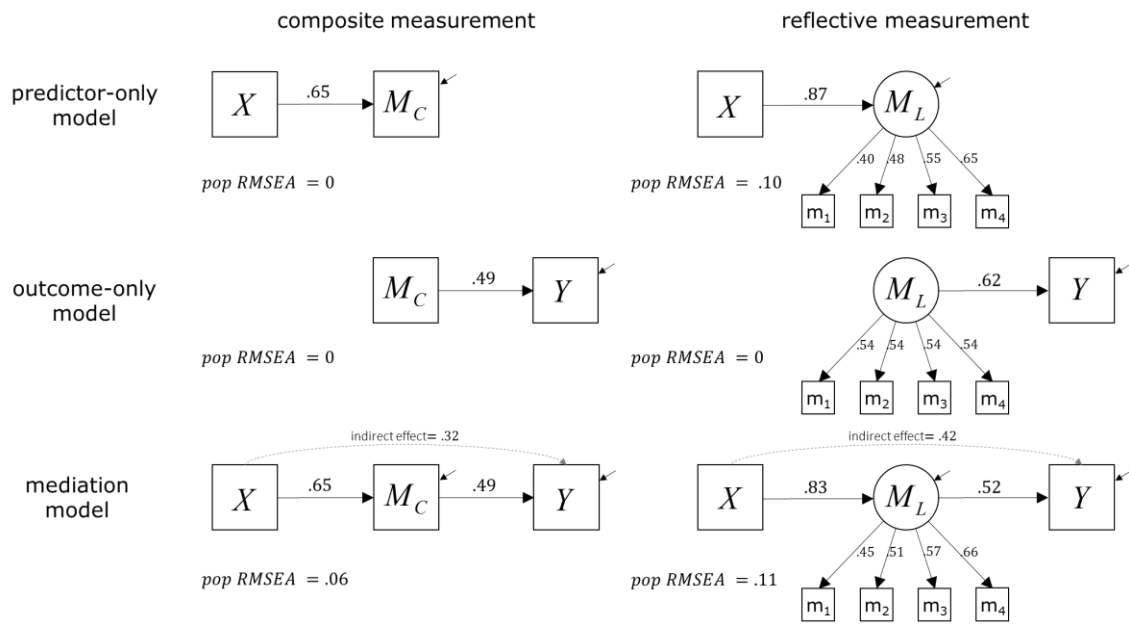


Figure 5. Fitted Models. In the composite measurement models (left column), $m_1 - m_4$

are summed to form M_c . In the common factor measurement models (right column), observed measures of $m_1 - m_4$ are modeled as reflective indicators of M_L . pop RMSEA is the population value of the root mean-squared error of approximation, which is a function of the minimum of the fit function and the model degrees of freedom:

$$\text{popRMSEA} = \sqrt{f / df} ; \text{ values closer to 0 reflect better fit.}$$

Examining Figure 5 reveals several notable results. First, the composite models produced overestimates of the a path and underestimates of the b path. The a path is overestimated because M_c includes the entirety of the effect of X on M (i.e., X is orthogonal to ζ , the unmeasured part of M), but the variance of M_c is smaller than the variance of M . Thus, the shared variance between X and M is a greater proportion of M_c than of M , resulting in an overestimate. To be precise, the covariance between X and M is scaled by $1 - \text{var}(\zeta)$ to produce

$\hat{a} = a / \sqrt{1 - \text{var}(\zeta)} = .6 / \sqrt{.75} = .69$, and this value is attenuated somewhat by the measurement error in M_c . The b path, in contrast, is underestimated because the covariance between M and Y must be multiplied by the covariance between M and M_c to get the shared variance between M_c and Y , yielding $\hat{b} = b(1 - \zeta) / \sqrt{1 - \text{var}(\zeta)} = b(.75) / \sqrt{.75} = .52$: this value is also attenuated somewhat due to measurement error in the measured causal indicators.

It is worth noting that this pattern of composite model results would be expected to change if, in the true population model, X were correlated with ζ . For example, if X is parental permissiveness and M is sexual media content, and if a relevant medium like television were not measured, then it would be reasonable to expect a relation between X and the unmeasured part of M . In that case, the estimated a path would get smaller.

A second notable finding is that the factor loadings across the three latent variable models are substantially different from each other. In the outcome-only model, all indicators have the same factor loading and the model returns perfect fit, because the total covariance among each pair of observed indicators is the same, as is the correlation between each indicator and Y . Fitting just this model would give the distinct impression that a reflective latent variable model is appropriate for these data. But when the predictor X is included in the structural model, suddenly

the factor loadings diverge from one another because the estimation procedure must optimize the fit both in terms of reproducing the covariances among indicators but also the covariances between each indicator and X , which differ. The optimal solution that is converged upon differs depending on whether Y is also in the model. This divergence in factor loadings means that the conceptual link between the proxy M_L and the construct differs across the three models, i.e., the meaning of the proxy variable changes: When X is in the model, M_L is mostly closely identified with its fourth indicator (i.e., the one that shares most variance with X), resulting in a highly inflated estimate of the a path. When Y is additionally in the model, X exerts somewhat less influence on the estimated factor loadings, producing a version of M_L that is still skewed toward X but less so. Without X in the model, M_L is equally related to each of its indicators. Importantly, in none of these models does M_L refer to the intended construct, M .

Finally, the indirect path from X to Y via M is more accurately estimated in the composite model than in the latent variable model. In the former, the indirect path is slightly attenuated due to measurement error, and in the latter, it is more dramatically elevated due to the false partitioning of variance in the causal indicators into shared and unique variance.

Parameter Sweep: Example 1

To get a sense of the variability of parameter estimates in composite and latent variable models that results from different true parameter values, we conducted a parameter sweep on a full range of possible parameter values for the model in Example 1, under a set of limiting constraints.

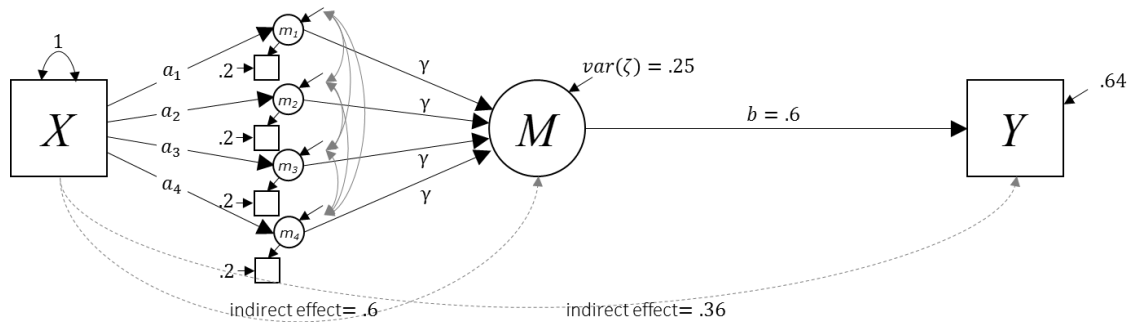


Figure 6. Population model for Example 1 parameter sweep. Each set of parameters was randomly drawn, subject to the following constraints: $a_1 - a_4$ path values were drawn

from $U(0, .8)$; the value of γ was chosen to ensure that the total indirect effect of X on M was .6; all bivariate correlations among $m_1 - m_4$ were equivalent and chosen so that the total variance of M would be 1.

We fit the 6 models in Figure 5 to 20,000 population covariance matrices. Each population was based on a randomly generated set of values of $a_1 - a_4$ (i.e., the paths from X to each causal indicator) from $U(0, .8)$. The value of γ (i.e., the path coefficient from each indicator to M , which was the same for all indicators) and the correlation between each pair of causal indicators (which was held equal for all pairs) was selected so that the total effect of X on the composite M was always $a = .6$ and the variances of $m_1 - m_4$ and M were 1. The residual covariance among each pair of causal indicators was computed based on the total correlation between each pair of indicators, subtracting the correlation implied by the $a_1 - a_4$ paths. Thus, though the marginal correlation among each pair of indicators was identical, their residual covariances differed as a function of their different $a_1 - a_4$ paths. The regression coefficient of Y on M was $b = .6$.

Method. R (R Development Core Team, 2016) was used to generate the population covariance matrices. The models shown in Figure 5 were fit to each population matrix using lavaan 0.5-23 (Rosseel, 2012). All R code for generating and fitting models is available at <https://osf.io/f8g4d/>. We evaluated the performance of the six models according to the accuracy of parameter estimates and model fit.

Accuracy. We examined the accuracy of the model-estimated a and b paths, which reflect the estimated effect of X on M , and of M on Y . Accuracy is assessed by plotting the observed estimates resulting from each model.

Model Fit. We examined model fit to determine whether larger bias in the estimated paths can be detected using fit statistics. We used the population RMSEA, $RMSEA = \sqrt{\hat{F}_{ML}/df}$, to indicate the degree of model misspecification in each generated population (Steiger, 1990; Steiger & Lind, 1980). The sample RMSEA is an unbiased and consistent estimator of this population value (Browne & Cudeck, 1993; McDonald, 1989).

Results. Figure 7 displays the standardized estimated $X \rightarrow M$ path (a) in the predictor-only model (top left) and the mediation model (bottom left), and the $M \rightarrow Y$ path (b) in the outcome-only model (top right) and the mediation model (bottom right). Each result is a point on the plot. These estimates are plotted as a function of the correlation between pairs of causal indicators in the true model. Purple points (which appear as a line) correspond to the composite model. Red points correspond to the reflective latent variable model, and the dark-to-light red color gradient represents the variability in the four $X \rightarrow m_i$ path values: results from models with more homogeneous path values are darker; those with more variable values are lighter.

Consider first the results for the composite model. As in Example 1, bias results from two factors, which can be seen here across the entire set of parameter values. First, as the correlation between causal indicators increases (i.e., moving left-to-right along the x-axis of each plot), the composite measure becomes more reliable because a greater proportion of its variance is due to these covariances relative to the measurement error in the observed variables. To be specific, as the correlation between causal indicators increases from .2 to .9, reliability of the observed composite increases from .86 to .94. This increase accounts for the small upward trend in estimated path values from left to right in each plot. Second, as explained in the previous section (Example 1), there is a constant trend for the a path to be overestimated and the b path to be underestimated, which is a function of the residual variance of M . In this parameter sweep, that residual variance was held constant at 25% of the total variance of M , so this bias was held constant.

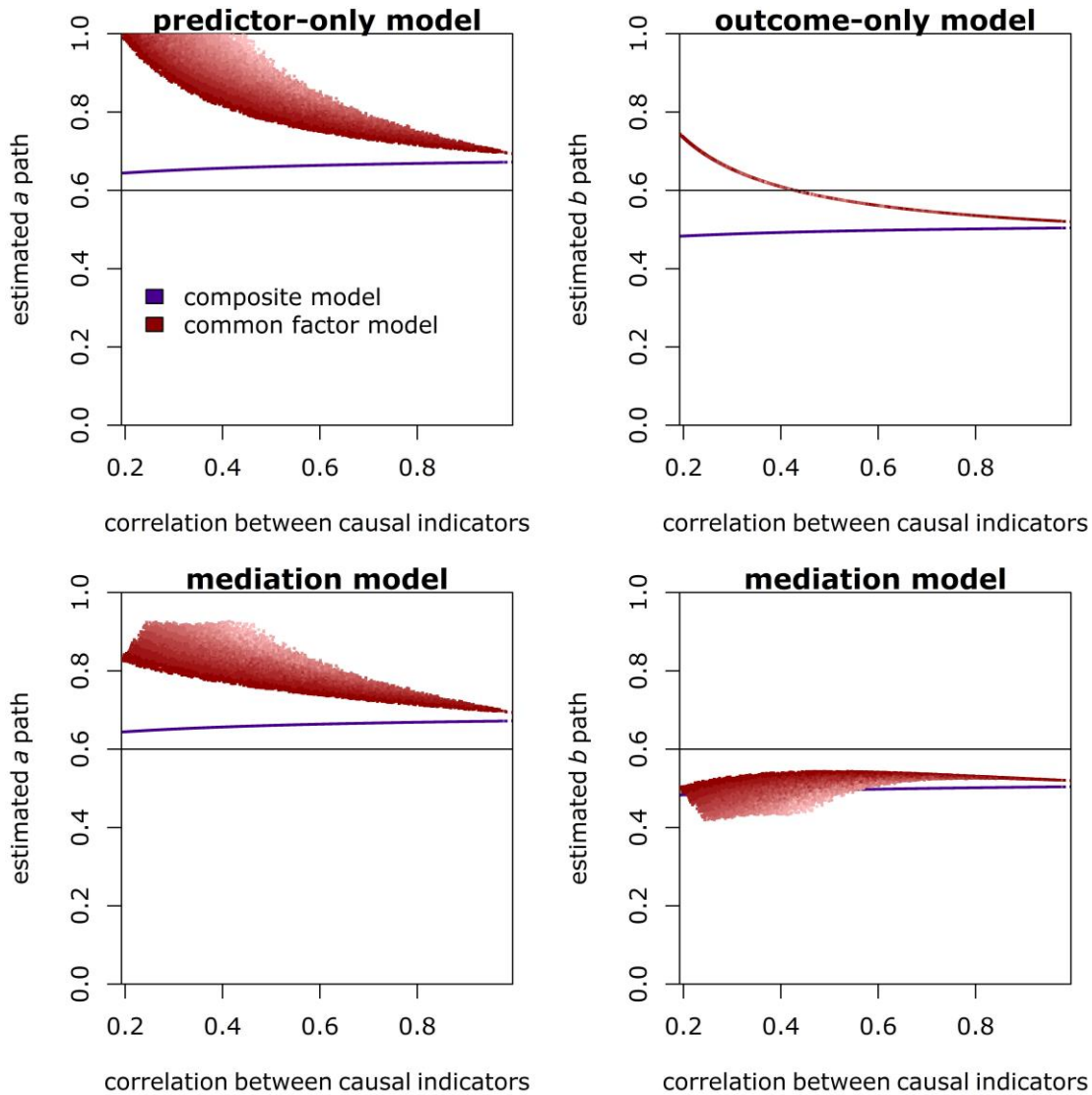


Figure 7. Example 1 parameter sweep results: Standardized path values. Purple points (which appear as a purple line) correspond to estimates from the composite model. Red points correspond to the common factor model. The dark-to-light red colour gradient indicates the variance in the four $X \rightarrow m_i$ path values: Darker indicates more homogeneity in these values. The black line at .6 indicates the parameter value in the true model.

Inspection of Figure 7 reveals three factors that influence bias in the structural coefficients of the reflective latent variable models. First, in the predictor-only and outcome-only models, the latent variable model always overestimates structural regression paths compared to the composite value. This overestimation is due to the misdisattenuation that we described earlier (see section “Bivariate Structural Correlations are Overestimated when the True Model is

Composite”). In the mediation model, the upward bias in the b path is ameliorated because the estimation routine searches for parameters that simultaneously account for the relations between $m_1 - m_4$ and X (as in the predictor-only model), $m_1 - m_4$ and Y (as in the outcome-only model), and now also between X and Y . As in the path model example given earlier, making the structural model more complex leads some paths that would have been overestimated in a simpler model to be underestimated when more variables are added to the model. Across the range of parameter values investigated in this sweep, higher estimates of a are associated with lower estimates of b ($r = -.9$).

Second, as the correlation among causal indicators increases (along the x-axis), a and b path coefficients from the common factor model become closer to the composite model estimates. Greater correlation among the indicators allows the common factor to capture more of these items’ variance (via larger factor loadings), making it a closer approximation of the composite score. In the predictor-only latent variable model, for example, higher average factor loadings were strongly correlated with less bias, $r = -.94$.

Third, as the variance of the set of $X \rightarrow m_i$ paths increases (i.e., points get lighter red), the common factor model increasingly suffers from differential item functioning; that is, a single path from X to M_L cannot capture the differential effects of X on the individual causal indicators, resulting in greater bias of the a path. This factor does not affect the bias of the b path in the outcome-only model. The b path is affected indirectly in the mediation model because its estimate is affected by the degree of bias in the a path.

Figure 8 displays the population RMSEA results for the predictor-only and mediation latent variable models. As in Example 1, in the outcome-only model, both the composite and latent variable models displayed perfect fit; the composite model also displayed perfect fit in the predictor-only model. The x-axis again represents the correlation between causal indicators. The dark-to-light red gradient again represents the variance of the $X \rightarrow m_i$ path values. These plots reveal that higher inter-item correlations are associated with both worse fit overall and greater variability in fit. This finding aligns with a literature showing that item reliability plays a large role in model fit, such that models with *less* reliable items tend to fit better (Miles & Shevlin, 2007; Savalei, 2012). Importantly, model fit is not a reliable indicator of the degree of bias in the

structural coefficients: Whereas higher correlations among the indicators leads to worse fit, it also tends to produce more accurate estimates (Figure 7). For example, in the predictor-only latent variable model, RMSEA was slightly negatively correlated with bias, $r = -.12$.

In the outcome-only model, fit was always perfect, despite a large degree of bias in the estimated path (in both the composite and latent variable models). However, greater variability in the $X \rightarrow m_i$ path values (displayed by lighter red points) leads to a greater violation of the proportionality constraint in the latent variable model, which produces both poorer estimates (as seen in Figure 7) and worse fit.

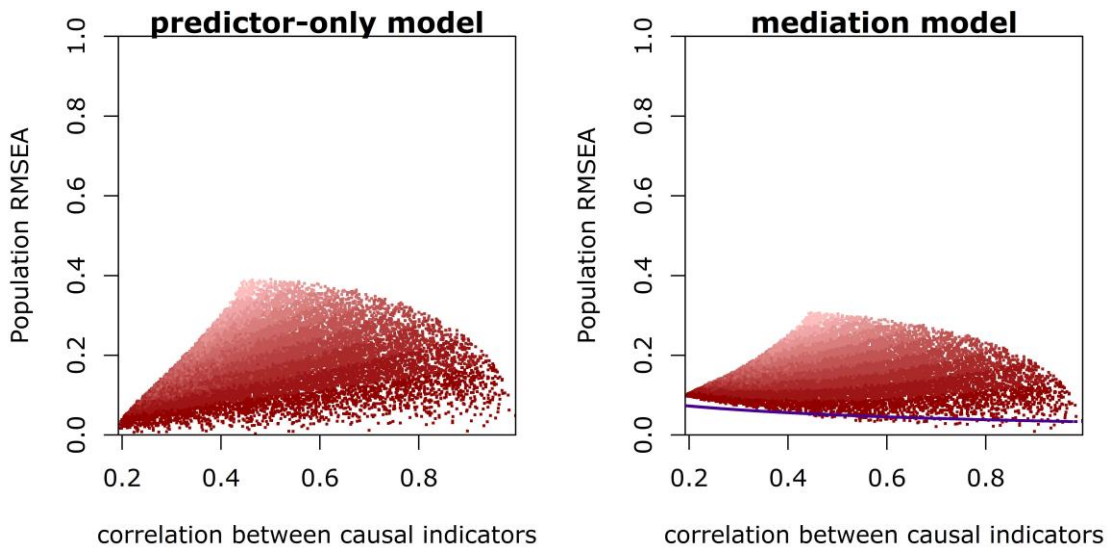


Figure 8. Example 1 parameter sweep results: Model fit. Red points correspond to the latent variable model and purple points correspond to the composite model. The composite predictor-only and outcome-only models have perfect fit due to having no constrained paths; the latent variable outcome-only models also have perfect fit due to the proportionality constraint being met in these models. The dark-to-light colour gradient indicates the variance in the four $X \rightarrow m_i$ path values: Darker points indicate more homogeneity in these values.

Example 2: MIMIC Model

Example 2 uses the same mediation model as Example 1, but the mediating variable M now has two causal indicators and two reflective indicators, making it a MIMIC model. For example, the effect of self-control on longevity might be mediated by lifestyle health, of which

causal indicators include vegetable intake and exercise and reflective indicators include self- and partner-reported perceived health. The true population model is depicted in Figure 9. M is caused by 2 imperfectly-measured causal indicators, m_1 and m_2 , which fully mediate the effect of X on M . In addition, there are two reflective indicators of M , m_3 and m_4 .

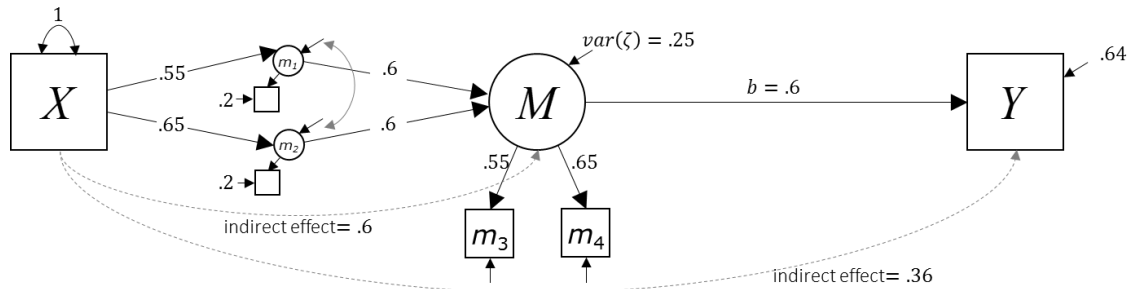


Figure 9. Example 2 true model. The total variance of each latent and observed variable in the model is 1, with the exception of residuals. The residual covariance between m_1 and m_2 is .34, and their total correlation is .5. The total indirect effect of X on M via m_1 and m_2 (depicted by the dotted line) is .6, and the indirect effect of X on Y is .36.

Figure 10 depicts the results of six models that were fit to the population covariance matrix implied by the model in Figure 9.

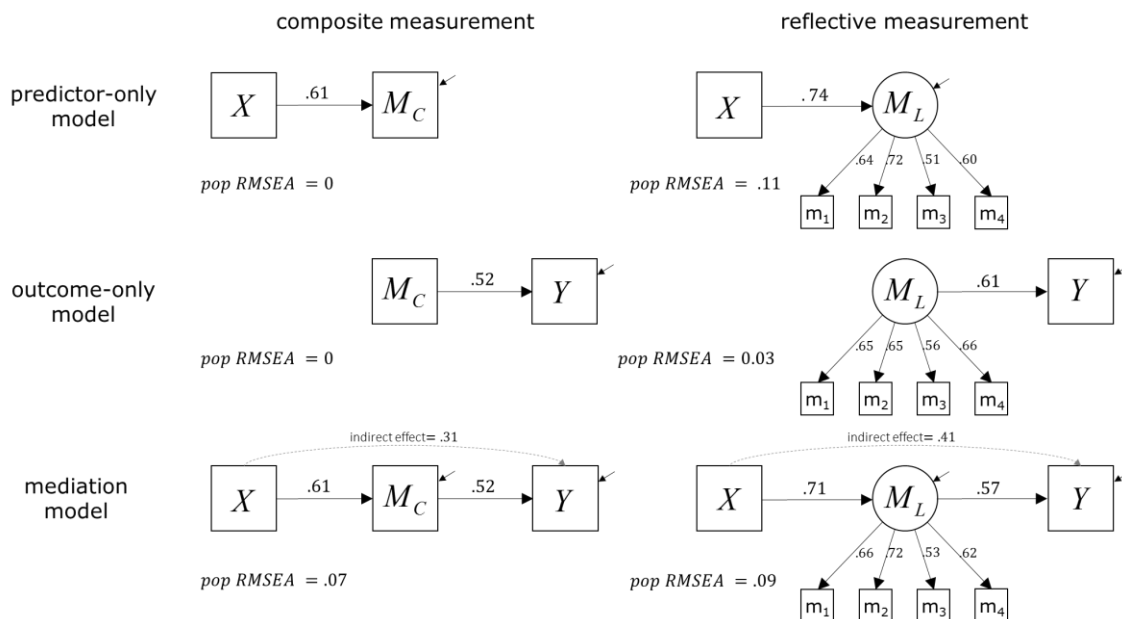


Figure 10. Fitted Models. In the composite measurement models (left column), observed measures of m_1 and m_2 are summed with m_3 and m_4 to form M_C . In the common factor measurement models (right column), the four observed variables are modeled as

reflective indicators of M_L . pop RMSEA is the population value of the root mean-squared error of approximation, which is a function of the minimum of the fit function and the model degrees of freedom: $\text{popRMSEA} = \sqrt{f / df}$; values closer to 0 reflect better fit.

In Example 1, the common factor measurement model perfectly fit the covariance structure of the 4 observed indicators. Misfit in the latent variable models arose because the latent variable measurement model was the wrong model for M , and thus it was unable to account for the relations between the indicators and the external variable X . In the present example, the CFA model no longer fits perfectly, because the covariances among the four indicators are no longer always equal.

As in Example 1, the MIMIC model results highlight some interesting disparities between the composite and reflective latent variable models. The composite model again shows positive bias in the a path and negative bias in the b path. This bias is again a result of the residual variance in M that is now present in the two reflective indicators of M but not in the causal indicators. There is also a tendency for both the a and b paths to be attenuated due to measurement error in all 4 of the observed indicators. The reflective models again show much higher path estimates of both the a and b paths, which is again a result of mis-disattenuation of the causal indicators. Finally, the three latent variable models again produce different factor loadings depending on the model, resulting in the latent variable carrying different meaning across the three representations.

We conducted a parameter sweep to accompany Example 2. The full description of this parameter sweep and its results can be found in the supplementary materials. Briefly, we found that both the composite and reflective latent variable models displayed less bias than they did when the true model had only causal indicators. We attribute this difference to the fact that the 2 truly reflective indicators in the model reflected *all* the variance of M , including its residual variance; thus, both the composite and latent variable proxies were able to capture more of the full variance of M . Otherwise, the results were quite similar to those of Example 1. As in Example 1, the reflective latent variable model estimates of the a path were typically more biased and substantially more variable than those of the composite model, and estimates of the b

path were strongly affected by whether the fitted model included X , Y , or both. The composite model estimates were not affected.

Example 3: Directed Network

Example 3 is a completely different kind of generating model for the construct M : M is no longer a unidimensional construct that is composed of or caused by its components; rather it is a network of interacting components. Such a network could assume many different forms, and in reality psychological networks are likely to contain feedback loops. For the sake of simplicity (and to be able to model the true model as a recursive SEM model), we consider the scenario in which M is a directed acyclic graph; that is, causal paths among the components are unidirectional and there are no feedback loops. Figure 11 displays the generating model.

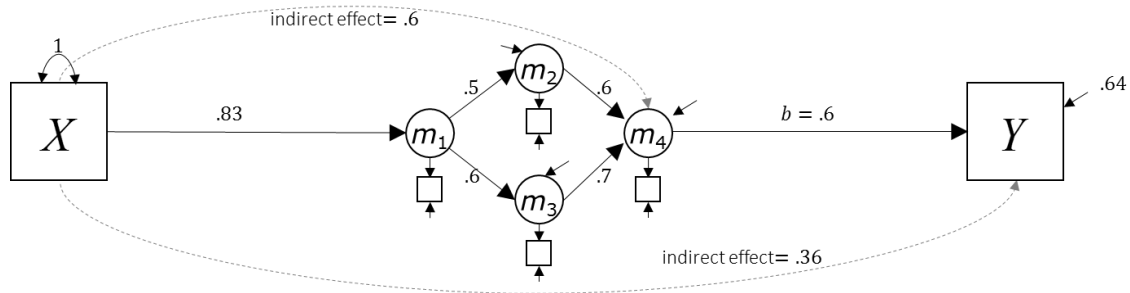


Figure 11. Example 3 true model. The residual variance of each observed indicator of $m_1 - m_4$ is .2 (not shown for clarity). The total variance of every variable in the model is 1, with the exception of residuals.

Figure 12 depicts the results of six models that were fit to the population covariance matrix implied by the model in Figure 11.

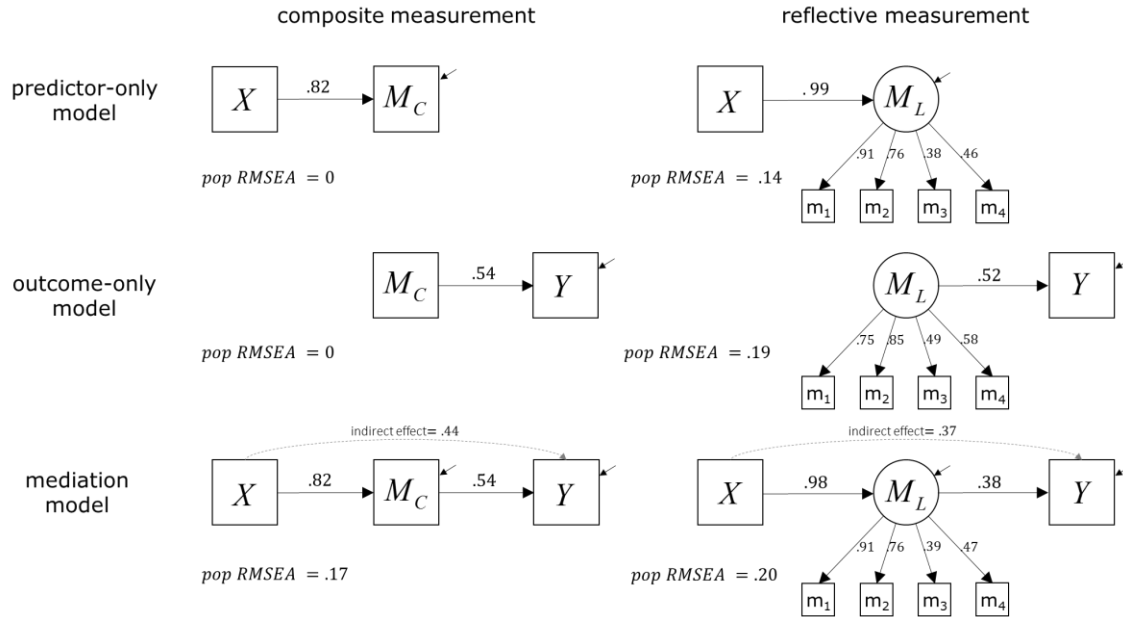


Figure 12. Fitted Models. In the composite measurement models (left column), $m_1 - m_4$ are summed to form M_C . In the common factor measurement models (right column), observed measures of $m_1 - m_4$ are modeled as reflective indicators of M_L . $pop RMSEA$ is the population value of the root mean-squared error of approximation, which is a function of the minimum of the fit function and the model degrees of freedom: $popRMSEA = \sqrt{f / df}$; values closer to 0 reflect better fit.

It is important to note that, as M is not a single unidimensional variable, there is no true value for the a or b paths. Thus, it is difficult to assess the degree of bias in these paths when the composite and reflective latent variable models are applied. It is nevertheless instructive to compare the two modeling frameworks to each other, and to compare across structural models. Doing this, we see that the path values differ substantially across the composite and latent variable models – in particular, the latent variable model estimates the a path to be an implausibly high .99 or .98 (recall that these are standardized path values) compared to .82 in the composite model. The value of the b path in the latent variable model is .52 when X is not included in the model but falls substantially to .38 when X is included. As in previous examples, this shift in structural parameter values is linked to a change in factor loadings, marking a change in the definition of the latent variable. As in all previous examples, the composite model path values do not change whether X or Y are included or not. This stability can be seen as an

advantage of the composite model: although its estimates may be biased, they appear to be far less volatile based on the contents of the structural model.

We conducted a parameter sweep to accompany Example 3. As the results show similar trends to the previous examples, we do not present them here but they are available in the supplementary materials.

Reanalysis of Empirical Data

Lee & Bull (2016) modeled the longitudinal relations among working memory, updating, and math performance in an effort to disentangle three hypotheses: (1) that working memory (WM) “predicts math performance more consistently than does prior math performance”; (2) that “the explanatory power of WM and updating is largely mediated by prior math achievement, and (3) that the relative importance of WM, updating, and prior math performance vary both with age and the domain of math under consideration.” To this end, they fit a cross-lagged panel model to examine the predictive relations between working memory, updating, and math performance.

Participants. The original study fit models to 4 cohorts of participants, each of which provided data in 4 consecutive school grades. Lee & Bull (2016)’s paper includes printed correlation matrices, standard deviations, and means for each variable in each cohort. For the sake of this example we used these summary statistics from just the second cohort, covering second grade to fifth grade ($N = 167$). The authors reported 14% missing data, so they used full information maximum likelihood to fit their models. Lacking access to the raw data, we simply fit models to the summary statistics. As a result, standard errors and test statistics from our re-analyses are not reliable, but parameter estimates should not be affected by these estimation differences.

Method. We fit 3 cross-lagged panel models to the data. In Model A, as in Lee & Bull (2016), Working Memory and Updating (WM/U) was fit as a latent factor at each time point with 3 reflective indicators that included 2 working memory tasks and a pictorial updating task. Numerical Operations (NO) was a single task score that represents math ability. Model A is identical to the model that Lee & Bull reported (based on syntax provided by the authors), except that we fit just one cohort rather than all 4, and we did not control for school cohort effects as we did not have access to the dummy-coded school variables that the authors used. As Lee & Bull (2016) did, we constrained the unstandardized cross-lagged paths from WM/U to NO (but not

from NO to WM/U) to be equal across all measurement occasions. Other details of model specification can be found in the R code at <https://osf.io/f8g4d/>. In Model B, the three indicators were summed to form a composite score that was entered into the model instead of the latent variable. Finally, in Model C, scores on the two working memory tasks were summed together to form a composite Working Memory (WM) variable, and the pictorial updating (PU) task was entered as a single proxy variable of a unique construct. We included Model C because it seemed in line with the authors' description of Working Memory and Updating as separate constructs (see Figure 13). Structural constraints that Lee & Bull imposed on Model A (e.g., orthogonal residuals between NO and WM/U in grades 1-3) were also imposed on Models B and C for the sake of conceptual similarity, despite resulting in substantially poorer fit.

Results. Figure 13 depicts the three models and their standardized parameter estimates. The latent variable model (Model A) displayed reasonable fit (though we note that several post-hoc modifications were imposed on the original model, so evidence for good fit should be interpreted cautiously): $\chi^2(93) = 137.82, p = .002, RMSEA = .05(.03, .07), CFI = .95, TLI = .93, SRMR = .06$.

The two composite-based models showed substantially worse fit, Model B:

$\chi^2(17) = 99.12, p < .001, RMSEA = .16(.13, .20), CFI = .85, TLI = .75, SRMR = .12$; Model C:

$\chi^2(37) = 123.2, p < .001, RMSEA = .11(.09, .14), CFI = .88, TLI = .78, SRMR = .10$.

Examining the standardized path values reveals some interesting discrepancies. First, the auto-regressive and cross-lagged paths from the latent WM/U variable to itself and to NO are substantially stronger than those for the composite WM/U, WM, and PU variables. This finding aligns with the examples presented earlier showing that the use of latent variables inflates bivariate relations relative to composite scores. In addition, Model A found no significant effects of NO on WM/U, which the authors interpreted to mean that “math performance did not predict WMU capacity at any point”, Models B and C found two-thirds of these paths to be significantly different from zero, and generally to be about the same strength as the paths from WM/U to NO.

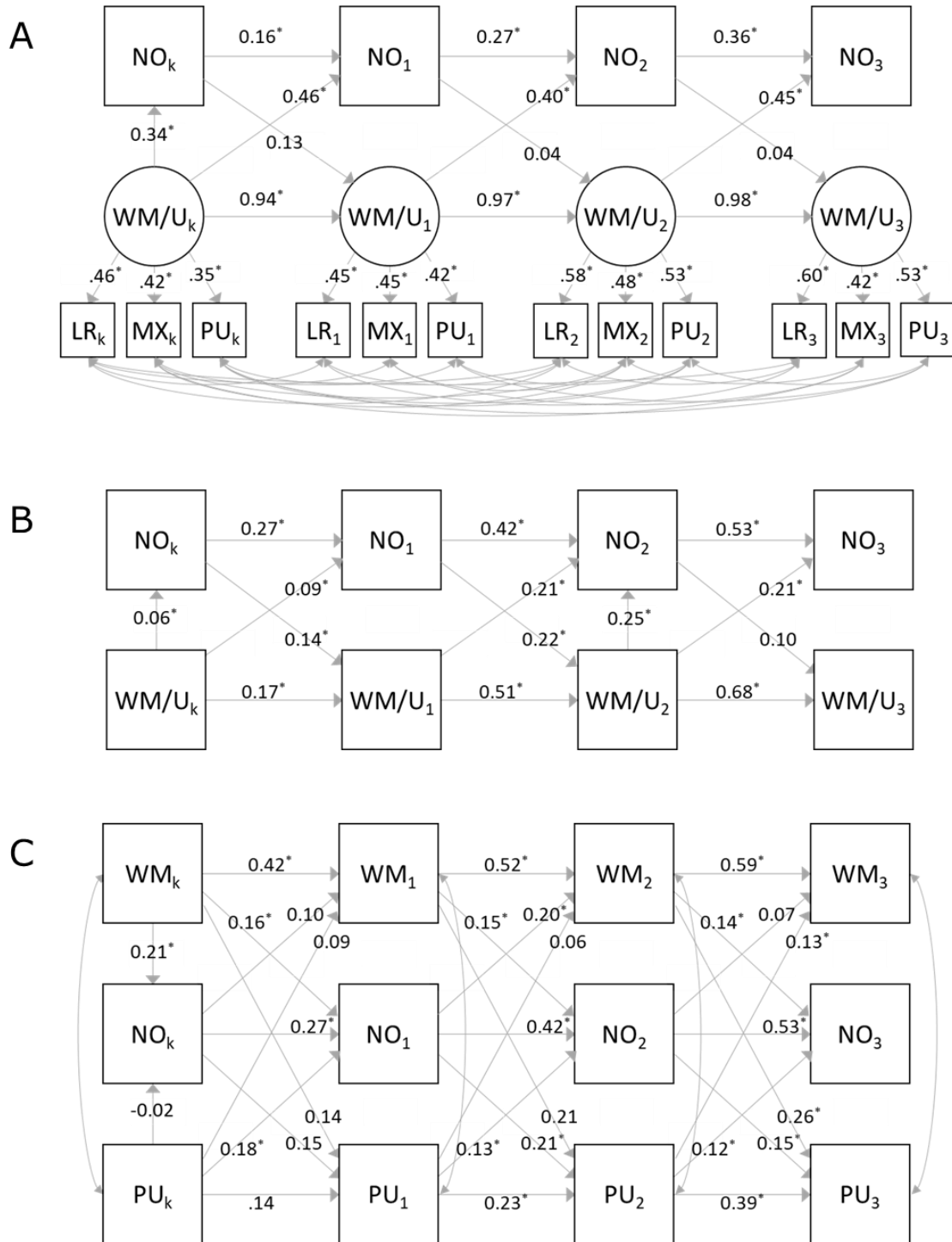


Figure 13. Three models fitted to data reported in Lee & Bull (2016). Subscripts refer to school grade during which data were collected. NO = Numerical Operations; LR = listening recall; MX = Mr. X; PU = Pictorial Updating. WM/U is an unweighted sum of LR + MX + PU. WM is an unweighted sum of LR + MX. Printed values are fully standardized estimates. $*$ = $p < .05$.

Discussion. We do not know what the best measurement model is for these data. What this re-analysis shows, however, is that the choice of measurement model seriously affects the interpretation of the statistical results in substantive terms. In the latent variable model, WM/U is characterized in terms of the shared variance among three cognitive tasks, two of which measure working memory, and the other updating capacity. As Lee & Bull (2016) describe, “On the measurement level, WM and updating are closely associated, ... but the two constructs are not conceptually identical (processing and recall vs. the selective replacement of information) and may contain different subcomponents.” By modeling WM/U as a single dimension of shared variance, the latent variable omits the unique “subcomponents” of WM and updating that could play a role in predicting other variables in the model. In contrast, if the WM/U construct is better construed as being multidimensional, then the composite model may be a better proxy for the construct, as it simply sums all the sources of variance that are contained within the indicators. Finally, we compared both these approaches with one in which WM and updating were modeled as proxies for two separate constructs, with each allowed to have unique effects on the outcome variable. All three models produced different parameter estimates, which would presumably have led the researchers to make different conclusions about the causal effects among these constructs over time. It is therefore important that the choice of model is based on informed theoretical considerations about the construct and the most plausible construct-item relations.

Discussion

Conventional wisdom in psychological measurement holds that there is no such thing as error-free measurement (e.g., Borsboom & Mellenbergh, 2002; Borsboom, 2005, 2008). Combined with the idea that *not* modeling measurement error when it is present leads to incorrect conclusions (Cole & Preacher, 2014), researchers may be under the impression that it is *always* better to use a measurement model than not. The current paper shows that this hypothesis is false: Applying an incorrect model for measurement error can be worse than not modeling measurement error at all, even when measurement error is present. This underscores the importance of thinking deeply about the structure of the model used to represent the relations between indicators and construct.

The present paper investigated the question of how the nature of the misspecification of the measurement error model affects the conclusions based on applying the model. First, we showed that when the true item-construct relation is composite rather than reflective, using a latent variable model incurs bias in the opposite direction than when a composite model is erroneously used in place of a reflective model. To extend this result to more complex models, we presented three examples of very different true population models, showing the results of fitting composite and latent variable models to each of them. These examples were accompanied by parameter sweeps that extend each example to a range of possible parameter values for each model. These parameter sweeps help to characterize the relationships between model characteristics (e.g., the strength of correlations among a set of observed indicators of a construct) and parameter bias as well as model fit. Finally, we fit composite and latent variable structural equation models to empirical data reported by Lee & Bull (2016). Our re-analysis found that the pattern of cross-lagged paths, including the relative strength of these paths and their statistical significance, depended strongly on the measurement model that was applied.

Our findings can be summarized as follows:

1. When a common factor model is fit to a set of items that truly represent components or causes of the construct, the common factor model produces overestimated values of structural correlations. We showed this result analytically when the true construct is a composite and in Example 1 when the indicators are truly causal indicators. This upward bias perpetuates through a structural model such that it can cause a complex pattern of both over- and under-estimated structural parameter estimates. This happens because the common factor model assumes that all variance not shared among indicators is unrelated to the construct. By excluding unique variance from the latent variable, shared variance at the structural level is estimated to be higher. When the common factor model is correct, the resulting estimates, disattenuated of unique variance, are more accurate. But when the true model is not a common factor model, “misdisattenuation” introduces bias.
2. The degree of bias resulting from the common factor model depends on aspects of the true generating model. For example, when the generating model was a causal indicator model or a MIMIC model, the lower the correlations were among the set of causal indicators, the more extreme the bias resulting from the common factor model.

3. The degree and direction of bias resulting from the common factor model depends on which variables are included in the structural part of the fitted model. We observed that when the fitted model included both a predictor and an outcome of the incorrectly modeled construct, estimated parameters were dramatically different than when only an outcome was modeled. This result may have important implications, for example, in cross-lagged panel modeling contexts where researchers may aim to evaluate whether an association between two variables over time is stronger in one direction than the other. In our empirical re-analysis of Lee & Bull (2016) we saw that the outcomes of such a cross-lagged panel model were strongly affected by the choice of measurement model.
4. Model fit is an unreliable indicator of the degree of bias in estimated parameters. It is possible to have a perfectly-fitting common factor model that nonetheless has highly biased parameter values compared to the true model, just as it is possible to have perfectly-fitting regression models that display attenuated path coefficients. Biased parameter estimates were sometimes but not always accompanied by poor fit, and the degree of misfit was seldom related to the degree of bias.

Common Factor vs. Composite Measurement

Our chief interest in this paper has been the performance of the common factor model, to address the question of whether, in the presence of measurement error, the common factor is able to adequately represent constructs under different population scenarios. For each example, we contrasted the fit of the inaccurate common factor measurement model with a composite model that was also an imperfect proxy for the construct. We chose to compare these two models because it is always possible to fit both models, regardless of what other variables are available to the researcher. Comparing the performance of these two incorrect models may help inform research practice when the true model is unknown.

We discovered that, in many of the examples we considered, parameter estimates resulting from composite models were less variable and less sensitive to parameter differences than those resulting from common factor models. While common factor models produced estimates that were highly sensitive to both the strength of correlations among the indicators and the variability in the relations between those indicators and other variables in the model, composite models were relatively robust to these differences. Thus, while the common factor

models sometimes produced more accurate parameter estimates compared to composite models, it was rare that the common factor model reliably produced more accurate estimates across all population values. In contrast, for several population models, the composite model was reliably more accurate across a range of parameter values.

In the models we examined, parameter estimates of the composite model did not change depending on whether the structural model included only a predictor, only an outcome, or both. This stability was in sharp contrast to the latent variable model parameter estimates, which changed dramatically depending on the presence of the predictor in the model. This finding suggests that composite-based models may be more robust to the features of the structural model than common factor models are. However, composite model estimates *are* also affected by the addition of more variables in models that are more complex than the ones we examined here (Cole and Preacher, 2014).

Though the composite model appears to have some advantages over the common cause model, composite-based estimates were also biased, sometimes strongly so. The primary factor that leads to bias in composite-based models is measurement error. But while latent-variable based models are able to correct for the presence of real measurement error, they over-correct (misdisattenuate) for false measurement error, that is, unique variance in observed variables that should rightly be considered part of the construct. Moreover, because the latent variable measurement model parameters (i.e., factor loadings and residual variances) are estimated concurrently with a larger structural model, the measurement model is sensitive to the variables and structure of the larger model (Burt, 1976). Each time different measurement model parameters are estimated, the link between the latent variable proxy and the construct under investigation changes. As Burt (1976) showed and as our findings reflect, this problem is worst when (a) the covariances among indicators are low, and (b) the covariances between individual indicators and other variables in the structural models are widely discrepant. For this reason, it has been argued that a composite-based approach (or a similar idea like principal component regression or PLS-SEM) should be preferred on the grounds that they produce results that are insensitive to elements of the structural model (Rigdon, 2012; Schönemann & Steiger, 1976) and that they should therefore be preferred. On the other hand, if a common-factor model plausibly

maps on to the true structure of the construct-item relations, then a common factor model should be preferred.

It is very difficult to judge which generating model is plausible based on the modeled data by themselves. As we have demonstrated in this paper, fit measures often do not indicate reliably which model most plausibly generated the data. This means that extra-statistical criteria are necessary to motivate the choice of model, and we think that there are considerable opportunities for future research that aims to highlight such criteria. While the full specification of substantive criteria for choosing a model for the indicators are beyond the scope of this paper, we do have a number of hypothesis on which dimensions may be relevant. The first mirrors Edwards and Bagozzi's (2000) thought experimental approach: the researcher would do well to ask the question of whether interventions on the construct should be expected to change the indicator values, or the other way around. The second is a novel question in response to the burgeoning of network analysis: Do interventions on one indicator plausibly induce changes in another indicator without mediation of a latent variable? For example, induced changes in sleep quality plausibly cause changes in fatigue, regardless of a latent variable like "depression". Third, and relatedly, an important question is whether there exists independent literature on causal relations among indicators. If such literature does exist (as it does for e.g. the sleep-fatigue link), it is more likely that relations among indicators are not fully attributable to a shared underlying common factor. Fourth, how plausible is the error model? The standard latent variable model assumes that errors behave in a highly specific way: they typically are modeled using normal distributions and are assumed to be independent of the targeted construct and other indicators. This behavior of error scores is best seen as an empirical hypothesis. If that empirical hypothesis is implausible for substantive or methodological reasons, it may be best to consider results based on latent variable approaches as reflecting this hypothesis rather than the data. Naturally, this list of relevant dimensions is tentative and plausibly incomplete, and considerable research should be directed at the question of how one should optimally choose between the different models.

Conclusions and Next Steps

We hope to have drawn attention to the problems that can arise from uncritical use of common factor measurement models for psychological constructs with the aim of correcting for

measurement error. We have shown that, regardless of measurement error, when the true relation between observed variables and the construct is not described by a reflective measurement model, using a common factor model can simultaneously correct for measurement error and lead to a dramatic mis-representation of the construct of interest. We contend that the correction for measurement error is not worth the potentially large cost to substantive interpretation.

Our findings highlight the importance of trying to identify an appropriate measurement model rather than relying on canned advice. The best way to distinguish between competing measurement models would be to test the causal relations among items and between construct and item via intervention – but doing so is often infeasible and frequently impossible². Thus, we suggest that a very good first step is for researchers to invoke theory and plausibility in describing the relations among items and the construct one wishes to measure. Articles in mainstream psychology journals appear to generally lack justification for the measurement models that are applied. When justification is given, it tends to be statistical justification in the form of a well-fitting confirmatory factor model. But as Studies 1 and 2 of the present paper show, a perfectly fitting confirmatory factor model can conceal a perfectly inappropriate measurement model (see also Hayduk, 2014).

Second, there is a clear need for a wider range of measurement models to embed in structural equation models. We have seen in this paper that the two standard options (common cause and composite models) do not always fit the bill. The use of causal indicator and MIMIC models is fraught with complications (e.g., see Howell et al., 2007, 2007a, replies by Bagozzi, 2007, and Bollen, 2007, as well as Bainter & Bollen, 2014, and the set of responses to that paper), and we know of no attempts to model such constructs as dependent variables in a structural equation model. Network models for psychological constructs are becoming more popular (e.g., Cramer et al., 2012), but again, to our knowledge, nobody has yet figured out how to embed these construct models within a larger structural model. Future work must focus on

² Some readers may protest that causality has nothing to do with the suitability of a common factor model; that common factors simply reflect shared variance among the indicators. In that case, to argue for a common factor model requires a case to be made that only the shared variance and not the unique variance of items is of interest. We can think of no rationale for such a case that does not hinge on a common cause (van Bork, Wijsen, & Rhemtulla, 2018)

expanding the kinds of measurement models that can be successfully embedded within structural models.

If psychology can achieve both these goals—if methodologists can make available a more complete range of measurement models and if researchers can be encouraged to spend effort to justify their choice of measurement model—we will have made much progress toward improving the inferences resulting from structural equation models.

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